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4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

MAY 2020

رمضان ١٣٢١

WEEK:-2 (PYTHON INTERMEDIATE)

WEDNESDAY 6
رمضان ١٣

PYTHON LISTS:-

mylist = ["apple", "banana", "cherry"]

LIST:-

→ Are used to store multiple ^{item} variables in single variable.

→ List are one of 4 built in types in Python used to store collection of data 3 are tuple, Set, Dictionary

Example:-

thislist = ["apple", "banana", "cherry"]

print(thislist)

Evening

LIST ITEMS:-

THURSDAY 7
رمضان ١٣

→ List items are ordered, changeable, and allow duplicate values.

→ The first item of index is [0]

ORDERED:-

→ It means that item have defined order that order will not change.

→ New item is place at end of list

CHANGEABLE:-

That we can change, add and remove item in list after it is created.

Evening

MAY 2020

رمضان ١٤٤١

MAY						
M	T	W	T	F	S	S
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

8 FRIDAY
رمضان ١٤ALLOW DUPLICATES:-

```
thislist = ["apple", "banana", "cherry", "apple", "cherry"]
print (thislist)
```

List can have items with same value

LIST LENGTH:-

```
thislist = ["apple", "banana", "cherry"]
print (len(thislist)) # output:- 3
```

LIST ITEMS:-

Same data type

list1 = ["apple", "banana", "cherry"]

list2 = [1, 2, 5, 7, 9]

list3 = [True, False, False]

Different data type

list1 = ["abc", 34, True, 40, "male"]

type() :-

mylist = ["apple", "banana", "cherry"]

print (type(mylist)) # output:- <class 'list'>

9 SATURDAY
رمضان ١٥10 SUNDAY
رمضان ١٦THE list() CONSTRUCTOR:-

To create a new list

```
thislist = list(("apple", "banana", "cherry"))
print (thislist)
```

PYTHON COLLECTION:-

	Ordered	changeable	unchangeable	unindexed	Duplicate member
List	✓	✓	✗	✗	✓
Tuple	✓	✗	✓	✗	✓
Set	✗	✗	✓	✓	✗
Dictionary	✓	✓	✗	✗	✗

Evening

MAY						
M	T	W	T	F	S	S
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

MAY 2020

١٥ رمضان ١٤٤١

MONDAY 11
١٦ رمضان

ACCESS LIST ITEMS :-

ACCESS ITEMS :-

Using index we can access the items.

```
thislist = ["apple", "banana", "cherry"]
```

```
print (thislist[1])
```

NEGATIVE INDEXING :-

-1 refer to last item

```
thislist = ["apple", "banana", "cherry"]
```

```
print (thislist[-1])
```

RANGE OF INDEXES :-

```
thislist = ["apple", "banana", "cherry", "orange", "kiwi", "lemon", "mango"]
```

```
print (thislist[2:5])
```

Output :- ["cherry", "orange", "kiwi"]

Evening

TUESDAY 12
١٧ رمضان

From start :- print (thislist[:4])

From end :- print (thislist[2:4])

Range of -ve indexes :- print (thislist[-4:-1])

CHECK IF ITEM EXISTS :-

```
thislist = ["apple", "banana", "cherry"]
```

```
if "apple" in thislist:
```

```
    print ("Yes, 'apple' is in fruit list")
```

CHANGE LIST ITEMS :-

CHANGE ITEM VALUE :-

```
thislist = ["apple", "banana", "cherry"]
```

```
thislist[1] = "blackcurrant"
```

```
print (thislist)
```

Evening

MAY 2020

رمضان ١٤٤١

MAY

M	T	W	T	F	S	S
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

13 WEDNESDAY
١٩ رمضانCHANGE RANGE OF ITEM VALUES :-

```

08 thislist = ["apple", "banana", "cherry", "orange", "kiwi", "mango"]
09 thislist[1:3] = ["blackcurrant", "watermelon"]
10 print(thislist)

```

INSERT ITEMS :-

```

12 thislist = ["apple", "banana", "cherry"]
01 thislist.insert(2, "watermelon")
02 print(thislist)

```

ADD LIST ITEMS :-APPEND ITEMS :-

```

05 thislist = ["apple", "banana", "cherry"]
Evening thislist.append("orange")
        print(thislist)

```

14 THURSDAY
٢٠ رمضانEXTEND LIST :-

```

08 thislist = ["apple", "banana", "cherry"]
09 tropical = ["mango", "pineapple", "papaya"]
10 thislist.extend(tropical)
11 print(thislist)

```

ADD ANY ITERABLE :-

```

01 thislist = ["apple", "banana", "cherry"]
02 thistuple = ("kiwi", "orange")
03 thistuple.thislist.extend(thistuple)
04 print(thislist)
05

```

Evening

MAY						
M	T	W	T	F	S	S
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

MAY 2020

رمضان ١٤٤١

FRIDAY 15
٢١ رمضان

REMOVE LIST ITEMS:-

REMOVE SPECIFIED ITEM:-

```

thislist = ["apple", "banana", "cherry"]
thislist.remove("banana")
print(thislist)
thislist.pop(1)
print(thislist)

```

Remove last item:- thislist.pop()

Remove first item:- del thislist[0]

Clear the list:- thislist.clear()

LOOP LISTS:-

LOOP THROUGH LIST:-

```

thislist = ["apple", "banana", "cherry"]
for x in thislist:
    print(x)

```

LOOP THROUGH INDEX NUMBER:-

```

thislist = ["apple", "banana", "cherry"]
for i in range(len(thislist)):
    print(thislist[i])

```

USING WHILE LOOP:-

```

thislist = ["apple", "banana", "cherry"]
i = 0
while i < len(thislist):
    print(thislist[i])
    i = i + 1

```

SUNDAY 17 SATURDAY 16
٢٣ رمضان ٢٢ رمضان

Evening

MAY 2020

رمضان ١٤٤١

MAY						
M	T	W	T	F	S	S
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

18 MONDAY ٢٢ رمضان LOOPING USING LIST COMPREHENSION:-

```
00 thislist = ["apple", "banana", "cherry"]
01 [print(x) for x in thislist]
```

10 LIST COMPREHENSION:-

11 Offers a shorter syntax when you want to create
12 new list based on value of existing list.

01 Example:-

```
02 fruits = ["apple", "banana", "cherry", "kiwi", "mango"]
03 newlist = []
04 for x in fruits:
05     if "a" in x:
06         newlist.append(x)
07 print(newlist)
```

19 TUESDAY ٢٣ رمضان In one line:-

```
08 newlist = [x for x in fruits if "a" in x]
09 print(newlist)
```

10 SYNTAX:-

11 newlist = [expression for item in iterable if condition == True]

12 Condition:-

```
01 newlist = [x for x in fruits if x != "apple"]
```

02 ITERABLE:-

```
03 newlist = [x for x in range(10)]
```

```
04 newlist = [x for x in range(10) if x < 5]
```

MAY						
M	T	W	T	F	S	S
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

MAY 2020

رمضان ١٤٤١

WEDNESDAY 20
رمضان ٢١EXPRESSION:-

```
newlist = [x.upper() for x in fruits]
```

```
newlist = ['hello' for x in fruits]
```

```
newlist = [x if x != "banana" else "orange" for x in fruits]
```

SORT LISTS:-ALPHANUMERICALLY:-

```
thislist = ["orange", "mango", "kiwi", "pineapple", "banana"]
```

```
thislist.sort()
```

```
print(thislist)
```

NUMERICALLY:-

```
thislist = [100, 50, 65, 82, 23]
```

```
thislist.sort()
```

```
print(thislist)
```

THURSDAY 21
رمضان ٢٢DESCENDING:-

```
thislist.sort(reverse = True)
```

```
print(thislist)
```

CUSTOMIZE SORT FUNCTION:-

```
def myfunc(n):
```

```
    return abs(n - 50)
```

```
thislist = [100, 50, 65, 82, 23]
```

```
thislist.sort(key = myfunc)
```

```
print(thislist)
```

Evening

MAY 2020

رمضان ١٤٤١

MAY

M	T	W	T	F	S	S
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

22 FRIDAY
٢٨ رمضانCASE INSENSITIVE SORT:-

thislist.sort()

print(thislist) # Output:- ["kiwi", "Orange", "banana", "cherry"]

thislist = ["banana", "Orange", "Kiwi", "cherry"]

thislist.sort(key = str.lower)

print(thislist)

REVERSE ORDER:-

thislist.reverse()

print(thislist)

COPY LISTS:- copy() method:-

thislist = ["apple", "banana", "cherry"]

Evening mylist = thislist.copy()

print(mylist)

23 SATURDAY
٢٩ رمضان24 SUNDAY
٣٠ رمضانLIST() method:-

mylist = list(thislist)

print(mylist)

SLICE OPERATOR:-

mylist = thislist[:]

print(mylist)

JOIN LIST:-JOIN TWO LIST:-

list1 = ["a", "b", "c"]

list2 = [1, 2, 3]

list3 = list1 + list2

Evening

print(list3)

MAY						
M	T	W	T	F	S	S
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

MAY 2020

شوال ۱۴۴۱

MONDAY 25
۱ شوالFor method:-

for x in list2:

list1.append(x)

print(list1)

Extend method:-

list1.extend(list2)

print(list1)

LIST METHODS:-METHODDESCRIPTION

append()

Add element at end of list

clear()

Remove all element from list

copy()

Return copy of list

count()

Return no. of element with specified value

extend()

Add element to end of current list

index()

Return index of 1st element with specified value

insert()

Add element at specified position

pop()

Remove element at specified position

remove()

Remove item with specified value.

reverse()

Reverse order of list

sort()

Sort the list.

TUESDAY 26
۲ شوال

Evening

MAY 2020

شوال ۱۴۴۱

MAY						
M	T	W	T	F	S	S
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4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

27 WEDNESDAY ۳ شوال PYTHON TUPLES :-

08 mytuple = ("apple", "banana", "cherry")

09 → Tuple item

10 → Ordered

11 → Unchangeable

12 → Allow Duplicates

01 UNPACK TUPLE :-

02 fruits = ("apple", "banana", "cherry", "strawberry", "raspberry")

03 (green, yellow, *red) = fruits.

04 print (green)

05 print (yellow)

Evening print (red)

Tuple methods :-

28 THURSDAY ۴ شوال → count() :- Return no. of items specified value occur in tuple.

09 → index() :- Search for specified value and return position where it was found.

1 PYTHON SET :-

12 myst = {"apple", "banana", "cherry"}

01 → Set items

02 → Unordered

03 → Unchangeable

04 → Duplicates not allowed

05

Evening

MAY						
M	T	W	T	F	S	S
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

MAY 2020

شوال ۱۴۴۱

FRIDAY 29
شوال ۱۴۴۱

PYTHON DICTIONARIES :-

```
thisdict = {
    "brand" = "Ford",
    "model" = "Mustang",
    "year" = 1964
}
```

- Dictionary items
- Ordered / Unordered
- Changeable
- Duplicate Not Allowed.

PYTHON FILE HANDLING :-

FILE HANDLING :-

SUNDAY 31 شوال
SATURDAY 30 شوال

- Key function for working with file in Python is open() function.
- Open() function have two parameters
 - ★ filename and mode.
- Four different methods to open a file.
 - ★ "r" = Read ★ "a" = Append ★ "w" = write ★ "x" = create
 - ★ "t" = Text ★ "b" = Binary

SYNTAX :-

```
f = open("demofile.txt")
f = open("demofile.txt", "rt")
```

Evening

JUNE 2020

شوال ۱۴۴۱

JUNE

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1 MONDAY ۸ شوال PYTHON READ FILE:-

```
f = open ("demofile.txt")
print (f.read())
```

```
f = open ("D:\\myfiles\\welcome.txt")
print (f.read())
```

01 USING WITH STATEMENT:-

```
with open ("demofile.txt") as f:
    print (f.read())
```

04 CLOSE FILES:-

```
f = open ("demofile.txt")
print (f.readline())
f.close()
```

2 TUESDAY ۹ شوال READ ONLY PARTS OF FILE:-

```
with open ("demofile.txt") as f:
    print (f.read(5))
```

10 READ LINE:-

```
with open ("demofile.txt") as f:
    print (f.readline())
```

01 In looping:-

```
with open ("demofile.txt") as f:
    for x in f:
        print (x)
```

Evening

JUNE

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15	16	17	18	19	20	21
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JUNE 2020

شوال ۱۴۴۱

WEDNESDAY 3
شوال ۱۴۴۱PYTHON WRITE / CREATE FILE :-

```
with open ("demofile.txt", "a") as f:
    f.write ("Now the file has more content!")
```

```
#open and read the file after appending
with open ("demofile.txt") as f:
    print (f.read())
```

OVERWRITE EXISTING CONTENT:-

```
with open ("demofile.txt", "w") as f:
    f.write ("Woops! I have delete the content!")
```

```
#open and read the file after the overwriting:
with open ("demofile.txt") as f:
    print (f.read())
```

Evening

CREATE A NEW FILE:-

```
f = open ("myfile.txt", "x")
```

THURSDAY 4
شوال ۱۴۴۱PYTHON DELETE FILES:-

```
import os
os.remove ("demofile.txt")
```

CHECK IF FILE EXIST:-

```
import os
if os.path.exists ("demofile.txt"):
    os.remove ("demofile.txt")
else:
```

```
    print ("The file does not exist")
```

DELETE FOLDER:-

```
import os
os.rmdir ("myfolder")
```

Evening